

Highly toughened PP/Rice husk charcoal composites modified EPDM Ethylene Propylene Diene Monomer (EPDM) with glycidyl methacrylate

Yaobin Wang

X.L. Deng

B.Y. Cao

H.P. Feng

J. Chen

P.D. Li

Liang Ren

renl@ccut.edu.cn

Changchun University of Technology <https://orcid.org/0000-0002-2502-2471>

Mingyao Zhang

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Abstract

In this article, a strategy is proposed to prepare highly toughened composites from polypropylene (PP), rice husk charcoal (RHC) and glycidyl methacrylate (GMA)-modified EPDM using melt blending and hot pressing. The structure of GMA-modified EPDM is confirmed by Fourier transform infrared spectroscopy (FTIR). The influences of the amounts of EPDM and GMA-modified EPDM on PP composites are investigated by virtue of mechanical properties testing, dimensional stability analysis, thermogravimetric analysis (TGA), X ray diffraction (XRD) and scanning electron microscopy (SEM). The results indicate that the incorporation of GMA-modified EPDM significantly improves the toughness of PP composites and compared to PRHC the impact strength of PRKEG10 is up to the value of 537J/m and increased by 555%, concurrently the tensile strength and modulus exhibit less decrease with the value of 8.22MPa and 353MPa, respectively. Shrinkage measurements shows the dimensional change rates of length, width, and height decrease from 3.2%, 2.9%, and 3.3% to 1.73%, 1.46%, and 1.93%, respectively, improving the dimensional stability of PP composites. SEM reveals that shear yield and cavitation during the loading process leads to the excellent toughness of PP composites. This strategy provides a novel route to fabricate high ductility rice husk charcoal-based PP composites and expand certain industrial application.

1. Introduction

Ever since its commercial debut in 1957^[1], Polypropylene (PP) resin has achieved remarkable rapid development, widely used in the automotive industry^[2,3], packaging field^[4-6], building materials^[7], electrical industry^[8,9], textile industry^[10,11], and so on, due to good comprehensive mechanical properties chemical stability, electrical insulation as well as easy processing and has become one of the world's largest consumed polymers with production soaring to 72.3million tons in 2020^[12]. At present, the PP products occur so ubiquitously that it is inconceivable for a market without PP. Because of the fact that there is still a long way to go in substituting petroleum-derived products with bio-based ones, saving PP consumption is presently a much more effective strategy to release the problems for alleviating the dependence on petroleum resources and reducing the pollution caused to the environment. In this context, PP composite products are emerging like mushrooms in production and daily life, which is of great significance for reducing the consumption of PP.

Biomass resources need to be given attention and incorporated into the economic cycle framework^[13]. It will become a key to solving environmental problems if it can be effectively converted and utilized due to its abundant sources. Rice husk charcoal (RHC) is industrial waste residue, a common biomass resource, which has a porous structure with high specific surface and SiO₂ content over 90% produced by burning rice husk as fuel in biomass power^[14-16]. Nowadays, much of RHC is discarded as waste without contributing to the development of a circular economy, also causing grievous environmental consequences and resource loss^[17-19]. Consequently, presenting a facile method for effectively enhancing utilization of RHC is of great practical significance, which not only generates considerable

economic benefits but also alleviates environmental crisis to a certain extent. To the best of the authors' knowledge, a large number of studies on the applications of RHC as a reinforcement agent for rubbers, plastics, cement, etc. have been issued^[20-23]. Zeng et al. have shown that modulus and tensile strength of DN-RHC/NR composites by modifying RHC with the rare earth coupling agent DN-8102 increased markedly^[23]. It is important to note that the incorporation of RHC into PP matrix unavoidably reveals the fact that simultaneously reinforcing and toughening of PP is essential to develop innovative PP-based composite materials with stiffness-to-toughness balance because of the tendency of PP to brittleness and low modulus^[24-26]. For this reason, a ternary composite consisting of polymer matrix, elastomer and filler have been developed as a promising material acquiring both good flexibility and high stiffness^[27]. Ethylene-propylene-diene monomer rubber (EPDM) is one of the most efficient tougheners to restore the proper toughness of PP/RHC composite yet it usually requires compatibilizers to improve the poor interfacial adhesion between them to obtain satisfactory mechanical properties^[28-30]. Khodabandelou et al. has published that the presence of PP-g-MA compatibilizer strengthens the adhesion between PP/EPDM blends resulting in a 140% increase in the impact strength of the composite^[31].

Aiming to fabricate a novel PP composite with stiffness-toughness balance and low

cost, this work attempts to control and regulate the phase morphology of EPDM in PP/RHC/EPDM or PP/RHC/EPDM-g-GMA ternary composites by incorporating two types of EPDM in different amounts at a fixed RHC content (20wt%). Subsequently, evolution of the phase morphology of these ternary systems upon gradual incorporation of EPDM is comprehensively studied by analyzing crystallization behavior, rheological performance, mechanical properties and dimensional stability as well as deformation microscopic mechanisms.

2. Experimental

2.1 Materials

PP (trade name: B8108) is supplied by Yanshan Petrochemical Co. (China). The used RHC (industrial grade) is obtained from Kaiyu Auto Parts Co. (China). EPDM is provided by PetroChina Jilin Petrochemical Company (China). Glycidyl methacrylate (GMA) and Dicumyl peroxide (DCP) is purchased from Aladdin Holdings Group (USA). The silane coupling agent (KH560) is purchased from Aladdin Holdings Group Co. (USA). Ethyl alcohol was obtained from Tianjin Fuyu Fine Chemical Co., Ltd (Tianjin, China). Ammonium Water provided by Beijing Chemical Works (Beijing China). Except for PP and RHC, all reagents are of analytical reagent grade and can be used without further purification.

2.2 Preparation of modified RHC

RHC is first washed with deionized water by filtration and then dried under vacuum at 80°C for 12h to remove residual water. The surface of RHC is modified by KH560, and the specific method is as follows: (1) An ethanol solution is prepared by mixing alcohol and water with the weight ratio of 90/10 and

adjusted the pH value of 8 by incorporation of ammonia water; (2) Add KH560 to the ethanol solution prepared in step (1) and stir for 1 hour at room temperature to make it fully hydrolyzed; (3) the dry RHC is added to the ethanol solution prepared in step (2) and stirred magnetically for 6 h; (4) Wash the solution with deionized water until neutral, and dry the precipitate in a vacuum oven at 80°C for 12 hours to obtain modified RHC, namely RHC@KH560.

2.3 Modification on EPDM

The surface of EPDM is modified by grafting GMA to enhance the interface compatibility between two phases and improve the performance of materials. EPDM-g-GMA copolymer is successfully fabricated by virtue of reactive blending EPDM, GMA and initiator DCP in internal mixer with the temperature of 180°C, the rotor speed of 60 r/min and blending time of 6 min. The reaction mechanism is shown in Figure 1.

2.4 Preparation of PP composites

PP composites consisting of PP/RHC@KH560/EPDM and PP/RHC@KH560/ EPDM-g-GMA are prepared by melt blending in internal mixer at the temperature of 210°C and rotor speed of 60rpm for 8min according to the formulations in Table 1. The flow chart of the composite preparation is shown in Figure 2.

Table 1. Components of PP-based composite materials

Sample	PP wt%	RHC wt%	RHC@KH560 wt%	EPDM wt%	EPDM-g-GMA wt%
PRHC	80	20	-	-	-
PRKE2	78	-	20	2	-
PRKE4	76	-	20	4	-
PRKE6	74	-	20	6	-
PRKE8	72	-	20	8	-
PRKE10	70	-	20	10	-
PRKEG2	78	-	20	-	2
PRKEG4	76	-	20	-	4
PRKEG6	74	-	20	-	6
PRKEG8	72	-	20	-	8
PRKEG10	70	-	20	-	10

2.5 Characterization and Measurements

2.5.1 FT-IR Test

The samples are scanned using the Fourier Transform Infrared Spectrometer (IS50, Nicholas, USA) with an ATR accessory and the range of wave number from 4000cm⁻¹ to 400cm⁻¹.

2.5.2 Purification of the EPDM-g-GMA

Add 50 ml of xylene to a three necked flask and heat it up to 150°C using an oil bath. Put 2 grams of EPDM-g-GMA into a three necked flask, heat and reflux for 2 hours, then quickly add 100 ml of acetone solution to precipitate it. The precipitate is vacuum dried at 60°C in a vacuum oven for 12 hours.

2.5.3 Determination of the grafting degree

The grafting degree of EPDM-g-GMA is determined by a reverse titration test. 0.5g of purified EPDM-g-GMA is dissolved in 50ml hot xylene, and then 100ml trichloroacetic acid is added while stirring to mix thoroughly and the experimental environment is in silicone oil at 140°C for 2h to ensure the ring opening reaction of the epoxy groups in the graft. KOH-ethanol solution is titrated in the above solution, using phenolphthalein as an indicator. The grafting degree (D_g) is calculated according to the following equation.

$$D_g(\%) = \frac{C \times (V_0 - V_1) \times 142.5}{1000 \times m} \times 100\% \quad (1)$$

where C is the concentration of the KOH-ethanol standard solution used for the calibration, unit mol/l; V_0 is the Volume of KOH-ethanol standard solution consumed at the end of the titration in the blank experiment, unit ml; V_1 is the volume of KOH-ethanol standard solution consumed at the end of the test experiment, unit ml; and m is the weight of sample.

2.5.4 Mechanical properties

Prior to testing, all specimens are kept at a constant temperature for 24hours to eliminate previous thermal stress. the impact test specimens are measured on an electronic cantilever beam impact tester (XJUD-5.5 type, Chengde Jinjian Testing Instruments Co., China) with an impact speed of 3.5m/s at room temperature, and each group sample is tested than more five times. The tensile strength of specimens is performed on an Instron tensile tester system (Instron-1121, USA) with a crosshead speed of 50mm/min at room temperature, and each component sample is repeated at least 3 times.

2.5.5 Shrinkage Measurements

Cuboid specimens with dimensions of 63.5mm×12.7mm×3.2mm (marking length) are prepared at 210°C, and dimensional change of the specimens is recorded after they are cooled to room temperature. The formula for calculating the rate of dimensional change of a sample is as follows:

$$S = \frac{L_1 - L_0}{L_0} \times 100\% \quad (2)$$

where S is rate of dimensional change of the specimen, unit %; L_0 is the initial length of the specimen, unit mm; L_1 is the length of the specimen after the test, unit mm.

2.5.6 Thermogravimetric Analysis (TGA)

The sample is analyzed using PYRIS-1 thermogravimetric analyzer, with a single sample mass of 2-4mg, a heating rate of 10°C/min, and a testing range of 30°C to 600°C.

2.5.7 Rheological property

The specimens are prepared using standard round molds with a thickness of 1mm and a diameter of 25mm. A rotational rheometer (AR2000, TA Instruments, USA) instrument is used for the linear viscoelastic properties of different samples in the angular frequency range from 0.1 to 100rad/s at 200°C.

2.5.8 X-ray diffraction (XRD)

The variation of crystal type of the PP-based composites is investigated using D/MAX 2000/PC instrument. The test range of scanning is from 5° to 50° and the scanning rate is 5°/min.

2.5.9 Scanning electron microscope (SEM)

The morphological studies of PP composites are investigated using a scanning electron microscope (JSM5600 type, JEOL Electronics Co., Japan). The all specimens with impact cross-section are sprayed with gold on the fractured surface before observation.

3. Results and discussion

3.1 FT-IR Analysis

GMA-modified EPDM is verified by comparing the FTIR spectra of purified EPDM-g-GMA and EPDM in Figure 3. It can be seen that the FTIR spectra of EPDM-g-GMA appears in a new absorption peak at 1736cm⁻¹ as compared to that of EPDM, which is consistent with the tensile vibration characteristic absorption peak of the C=O bond in GMA^[32], and thereby the evidence preliminarily confirm that GMA is successfully grafted onto EPDM to form EPDM-g-GMA copolymer.

3.2 Grafting degree of EPDM-g-GMA

Figure 4 shows that the influence of amount of GMA on grafting degree of EPDM-g-GMA. It is found that the grafting degree of EPDM-g-GMA is directly proportional to the amount of GMA until an 8wt%

concentration of GMA, at which point the grafting degree reaches a maximum value of 5.2%. On the contrary, an amount of GMA exceeding 8% does not further contribute to the grafting degree; instead, excessive GMA leads to a decrease in the grafting degree. This is because the number of grafting points contained in the EPDM molecular chain is fixed. Before reaching the saturation point, increasing the amount of GMA can increase the collision probability between GMA and the grafting points on the EPDM molecular chain, thereby increasing the grafting rate^[33-34]. However, when GMA is in excess, there are not enough grafting points on the EPDM to react with the GMA, leading to spontaneous polymerization of the GMA monomer under the influence of an initiator. This hinders the grafting reaction, causing the grafting degree of EPDM to slowly decline.

3.3 Mechanical properties Analysis

The change in impact strength of PP composites with varying amount of unmodified and modified EPDM, at a fixed RHC addition of 20%, is shown in Figures 5. Figure 5 reveals that the impact strength of both composites significantly improves with an increase in the amount of EPDM, with the 10% unmodified and modified EPDM performing best. They respectively enhance the impact strength of the composites from 82J/m to 430J/m and 537 J/m, marking an increase of approximately 424% and 555%, respectively. This remarkable improvement can be attributed to the localization of the RHC around the EPDM rubbery phase, which leads to an overlap of the stress fields surrounding the EPDM rubber and the PP matrix^[35]. To put it simply, the incorporation of rigid RHC into PP matrix further increases the heterogeneity of the material and makes it more prone to stress concentration at the phase interface, resulting in the presence of a large number of "nuclei" that can form crack. Meanwhile, the presence of rubber component tends to crazing and shear yielding so as to a toughening effect. Furthermore, GMA-modified EPDM is more effective in promoting the impact properties of composites compared with pure EPDM^[36]. This is probably due to the fact that GMA-modified EPDM becomes an elastomer with surface polarity and could be more efficiently dispersed in the PP matrix, which would increase craze and shear yielding effectively in PP matrix, exhausting impact energy massively^[37]. However, EPDM promotes an increase in internal flexible molecular chains, which are more prone to deformation under the action of external forces, and this inevitably results in a lower modulus^[38]. As shown in Figure 6, the decreasing stiffness is caused by the increasing amount of EPDM.

3.4 Shrinkage Analysis

Figure 7 displays the test results of the dimensional change rate of PRHC, PRK and PRKEG2. The molding shrinkage of the materials can be calculated by Equation 2. It is noticed that the incorporation of RHC@KH560 and EPDM-g-GMA into PP matrix remarkably reduces the dimensional change rate in length from 3.2% to 2.9% and 1.73% respectively, indicating improved dimensional stability. The same trend is observed for width and height. The effect of EPDM-g-GMA amounts on the dimensional stability of PP composites is given in Figure 8. The analysis indicates that the molding shrinkage of PP composite decreases significantly with increasing the amount of EPDM-g-GMA. Owing to the chain

entanglement arising between GMA molecular chain and PP due to the presence of GMA molecular chain in composites^[39], PRKEG has higher dimensional stability as compared to PRK and PRHC.

3.5 Thermogravimetric analysis

Figure 9 presents the thermogravimetric curve of PP and PP composites. It has greatly improved the thermal stability of PP composites due to the addition of RHC. This can be attributed to the strong interaction between RHC and PP, which can restrict the movement of polymer molecular chains, and the RHC can hinder the heat transfer and the volatilization of the decomposition products^[40]. PRK shows an increase of thermal decomposition temperature compared to PRHC, reaching 514°C, owing to the enhanced compatibility of PP composites resulting from the grafting of KH560 onto RHC, PRK exhibits a higher degree of thermal stability as compared to PRHC^[41]. The addition of GMA-modified EPDM will reduce its stability to PRK^[42], when addition amount reaches 10%, the thermal decomposition temperature decreases from 514°C to 426°C, but there is still a 51°C increasing compared with PP.

3.6 Rheological behavior of PP composites

The processing properties that are an important performance indicator in practical applications for polymeric materials can be characterized by rheological performance tests. The curves of energy storage modulus (G'), loss modulus (G'') and complex viscosity versus angular frequency for different amounts of elastomeric composites are shown in Figures 10 and 11.

As can be seen from the figure, a relatively markedly increase in G' and G'' occurred upon adding EPDM at low angular frequency ($<1\text{rad/s}$), exhibiting a positive proportional function relationship, while the effect of EPDM rubbery phase on the G' and G'' at high angular frequencies ($>1\text{rad/s}$) is hardly found. The reason is probably that cross-linking occurs between the PP matrix molecular chains and the added EPDM molecular chains during processing. At low angular frequencies, the cross-linking between the molecular chains facilitate the chain entanglement degree of the composite because the speed of movement of the entangled molecular chains is greater than the speed of disentanglement under the action of external forces, and then an increment in the number of entangled molecular chains in composites caused by continuous increment of EPDM content form a network structure, which appears in the enhancement in the G' and G'' as a result of decreasing fluidity of the molecular chains and increasing viscosity of the system^[43]. By contrast, the weak interaction forces between the molecular chains of the polypropylene matrix and the EPDM rubber molecular chains are unlikely to form such a network structure at high angular frequencies, and hence the G' and G'' are not significantly changed by the addition of elastomer^[44]. In summary, the increase in melt strength of the composites due to the EPDM rubber led to a further improvement in the processing formability of the material, which is consistent with the previous change in dimensional stability. Figure 10(c) and 11(c) shows the variation curve of complex viscosity with angular frequency for the composites. The complex viscosity (η^*) of PP composites are significantly enhanced by the addition of EPDM at low frequency regions. The increase of EPDM content tends to form more mesh structure between the two phases of PP and EPDM, making

the η^* of the composite increase, and the spatial site resistance of polymer molecules becomes larger and the molecular chain movement is difficult, which is characterized by the gradual increase of viscosity of the material in the molten state. It is worth noting that the G' and G'' along with η^* of PRKEG composites are higher than those of PRKE composites due to the presence of GMA which strengthens the interaction forces between the PP and EPDM molecular chains.

3.7 X-ray diffraction analysis

The influence of EPDM on the material composition, microstructure, and morphological information of PP composites can be understood in Figure 12. It can be observed from Figure 12 that five main diffraction peaks appear at $2\theta=14.7^\circ$, 17.5° , 18.3° , 22.5° and 26.1° for different PP composites, respectively. We can learn that the incorporation of EPDM and RHC into the PP matrix does not affect the crystalline shape of composites because there is almost no difference in the position of the diffraction peaks and their area^[45-46]. GMA is referred as a compatibilizer that improves the incompatible phenomenon between PP and EPDM, and is proven to heighten the compatibility without changing the crystallization structure of the PP composites.

3.8 SEM Analysis

Figure 13 illustrates the microscopic morphology in the impact section of different composites. In Figure 13a, the fracture surface of the PRHC with many holes and relatively smooth indicates that the RHC is detached from the matrix under the impact force and failed to withstand the impact force together with the matrix, thus displaying a brittle mode of fracture with low toughness. On the fracture surface of PRKEG4, a large number of small hollows left on the surface and the coarse structure of the deformed PP matrix can be observed (Figure 13b). This is attributed to the cavitation of the EPDM-g-GMA particles encapsulating the RHC under punching load pressure, which allows the matrix polymer to undergo a shear yielding process between the two cavities, thus greatly increasing the fracture toughness^[47-48]. The obviously rough fracture surface is due to the formation of pores and relatively strong shear yielding in the polypropylene matrix, as shown in Figure 13c. Internal air pockets and/or peeling of EPDM rubber particles from the PP substrate are likely to be the causal factors for the formation of these voids. In addition, RHC particles may detach and/or pull out from the continuous matrix under external effects, which can also trigger some cavitation phenomenon^[42]. Compared with the PRKEG6 composite, the shear yield in the PRKEG10 matrix is much more abundant, and the degree of plastic growth of micropores is also much more significant. Furthermore, on the fracture surface of the PRKEG10 composite, the formation of irregular stripes consisting of highly deformed substrate material can be observed^[49], which suggests that PRKEG10 composite exhibits more toughness relative to PRKEG6 due to this mechanical deformation that consumes more energy during impact fracture.

4. Conclusions

EPDM-g-GMA makes impact resistance increases by 555% at 10 wt%, indicating EPDM-g-GMA is selected to be added to achieve the purpose of toughening.

Shrinkage measurements show the dimensional stability is improved in length, width, and height from 3.2%, 2.9%, and 3.3% to 1.73%, 1.46%, and 1.93%, respectively.

The finding of TGA indicates the thermal decomposition temperature of PRKEG10 composite increases from 375°C to 426°C, which enhances the thermal stability.

The rheological properties tests reveal that GMA-modified EPDM plays an important role in easy processability.

SEM result shows the emergence and increase in shear yielding and cavitation promotes the improvement of composite toughness.

Declarations

Notes

The authors declare no competing financial interest.

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Figures

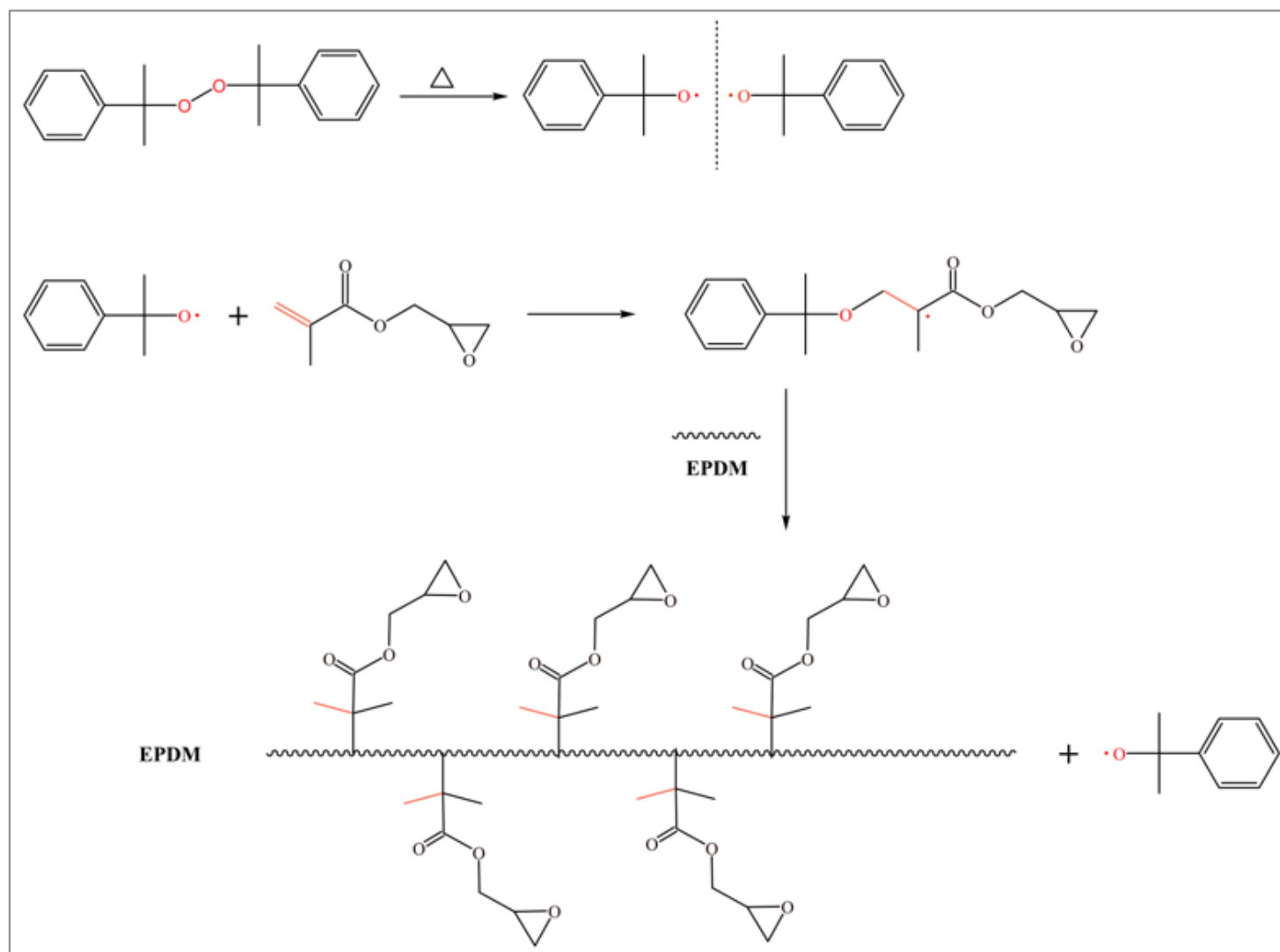


Figure 1

The reaction mechanism of EPDM-g-GMA

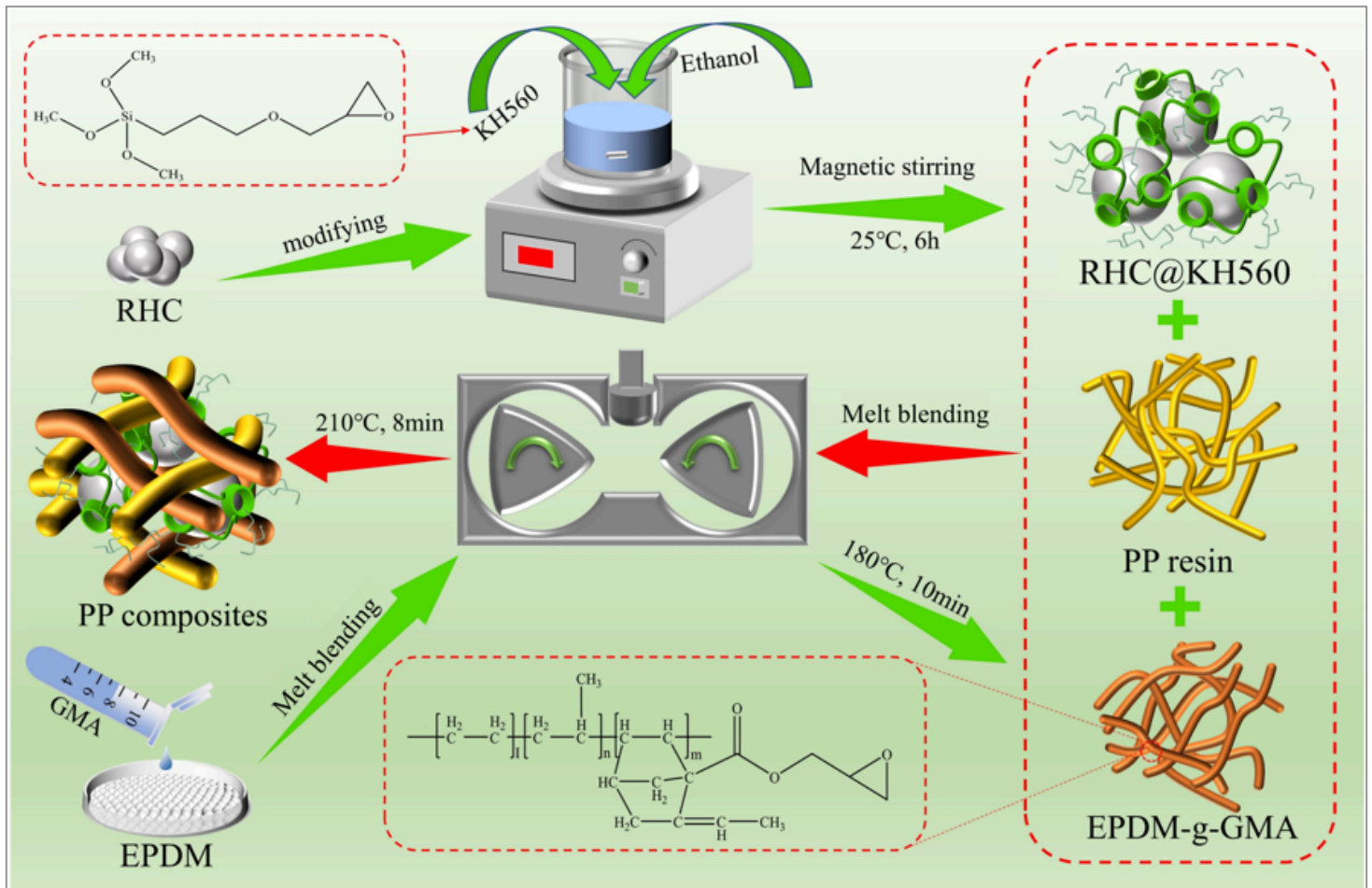


Figure 2

The flow chart of PP composite preparation

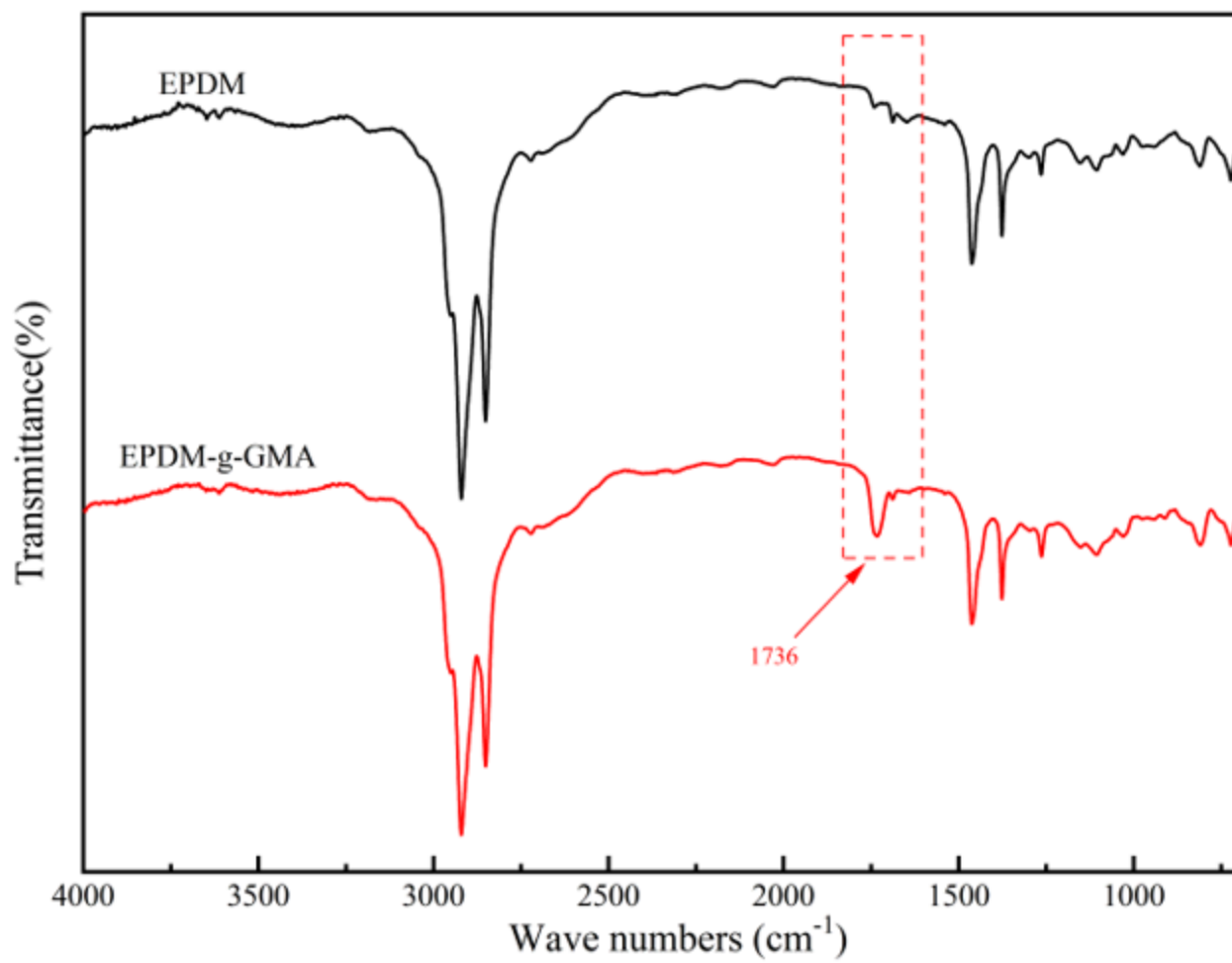


Figure 3

FT-IR spectrum of the EPDM and EPDM-g-GMA

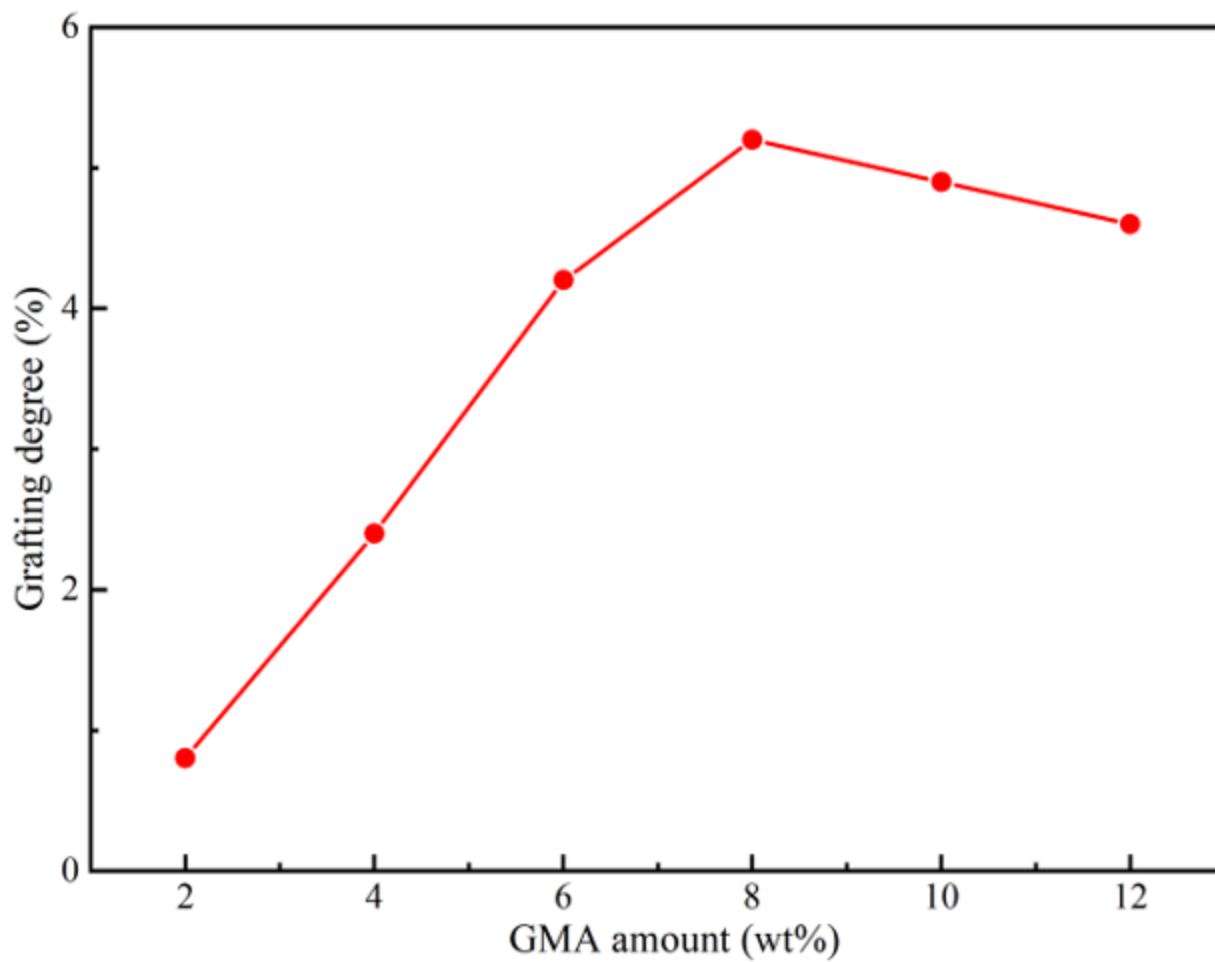


Figure 4

Effect of GMA amount on grafting degree of EPDM-g-GMA

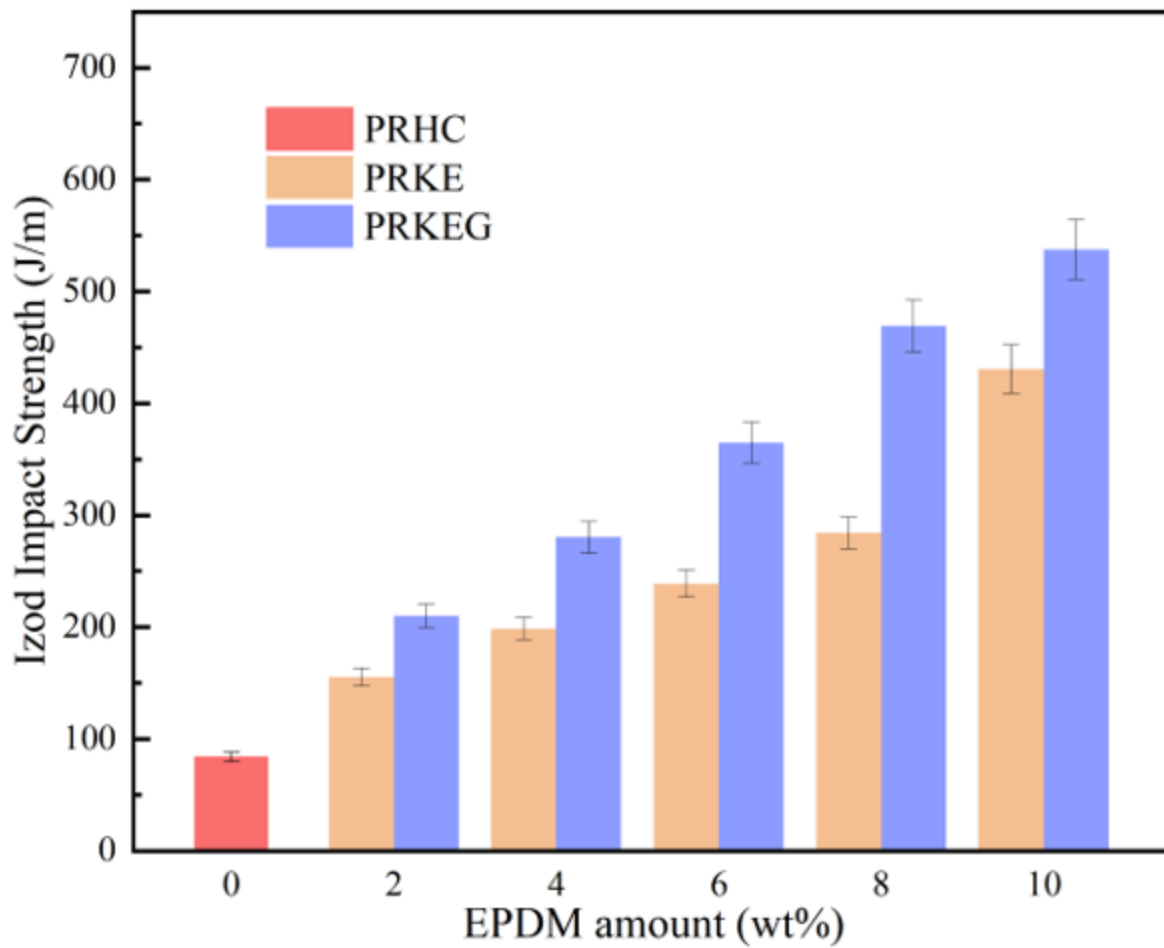


Figure 5

Izod Impact Strength of PP composites

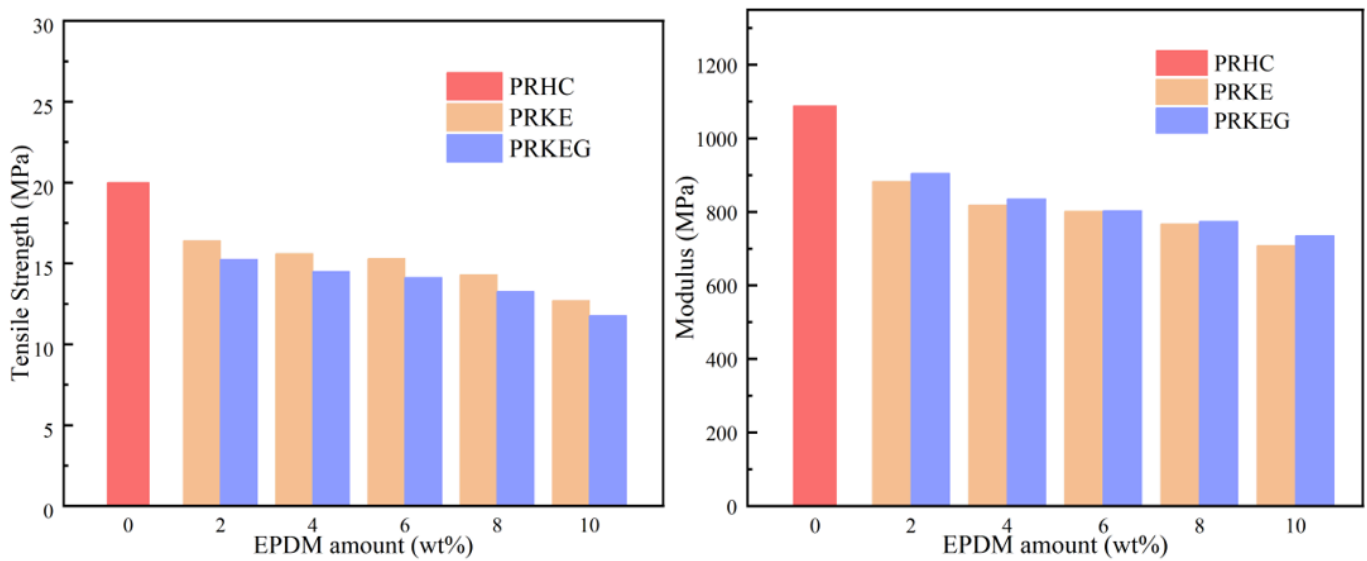


Figure 6

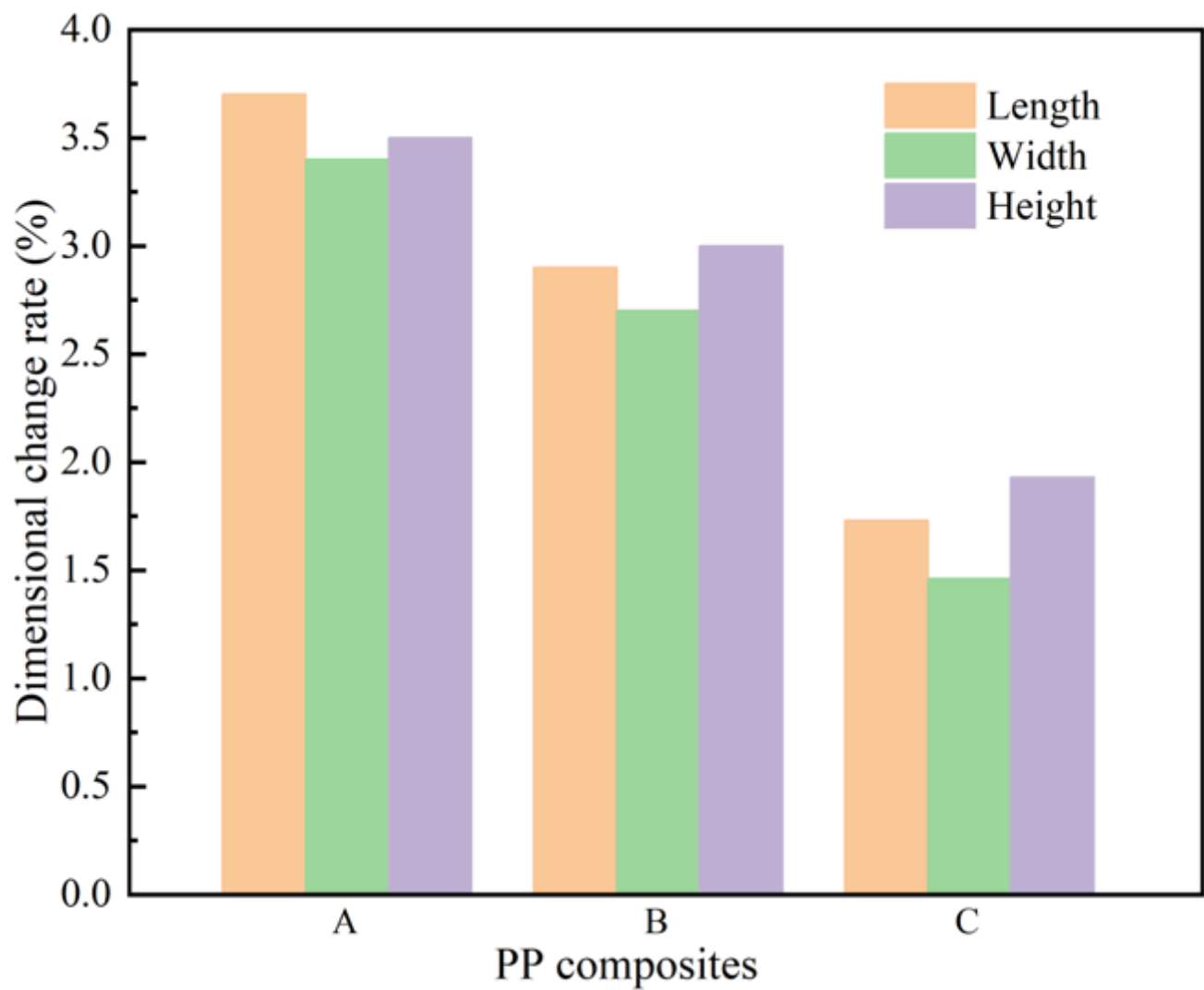


Figure 7
Dimensional change rate of PP composites: (A) PRHC; (B) PRK;
(C) PRKEG2

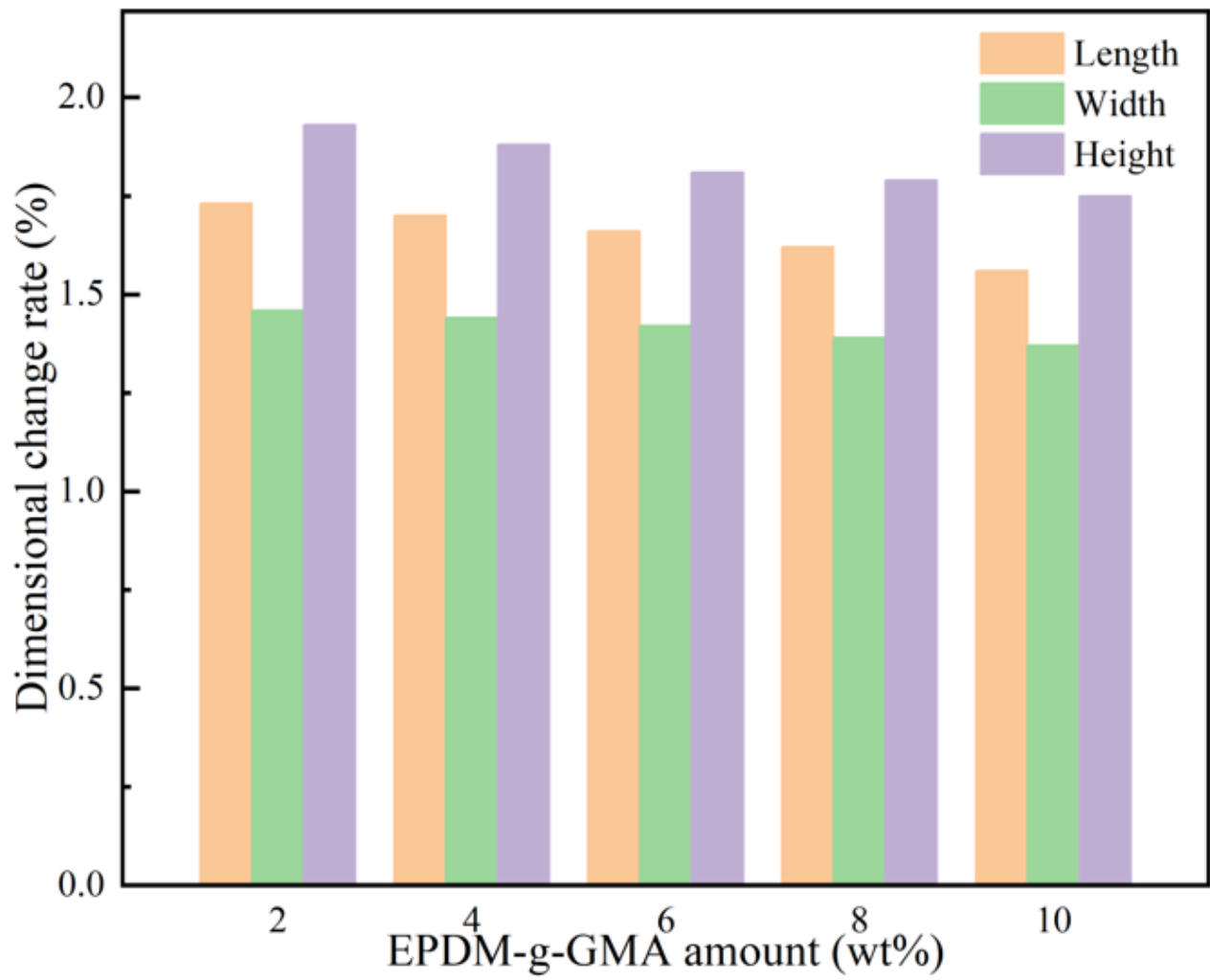


Figure 8

Dimensional change rate as a function of EPDM-g-GMA amount for PP composites

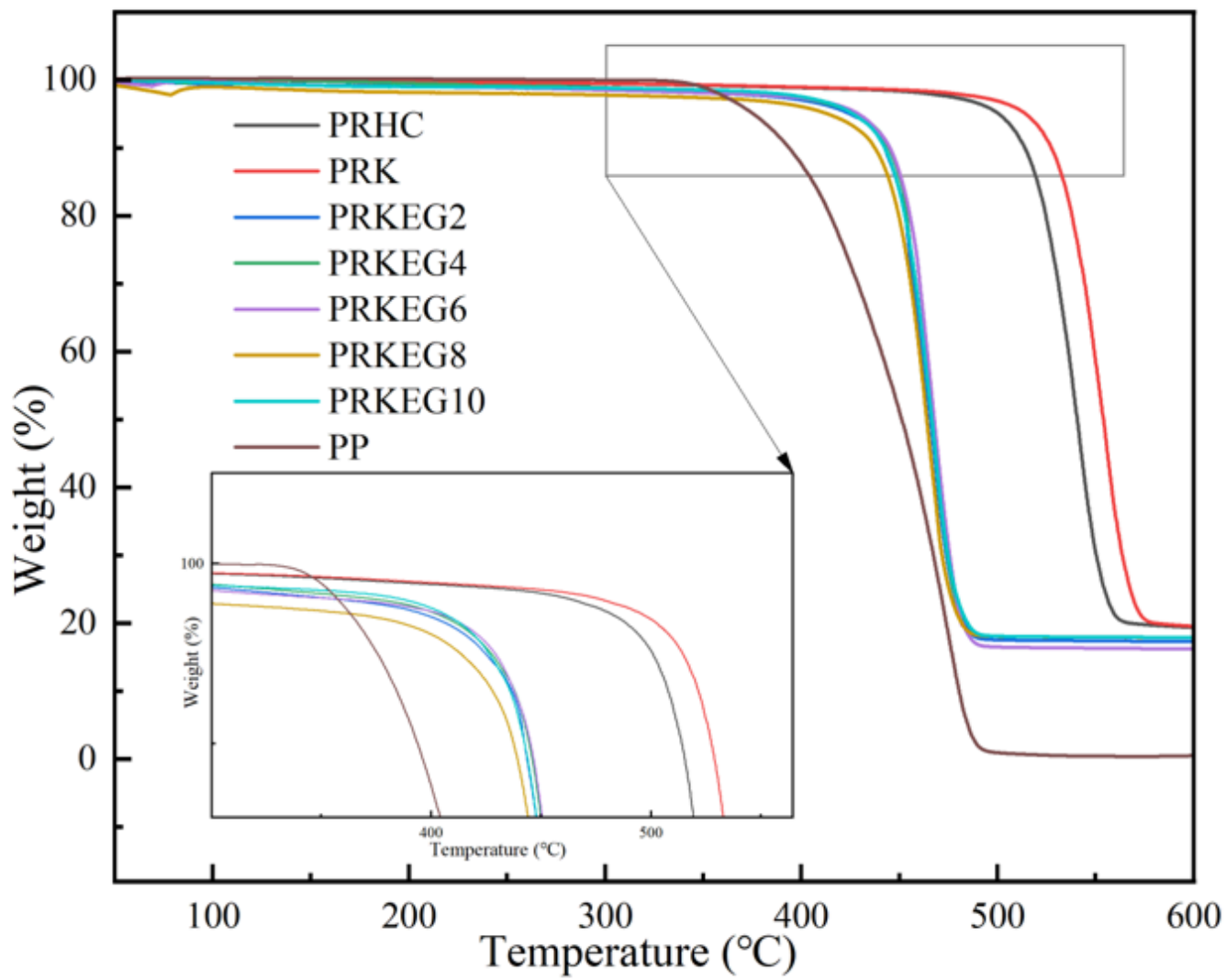


Figure 9

TGA curves of PP and PP composites

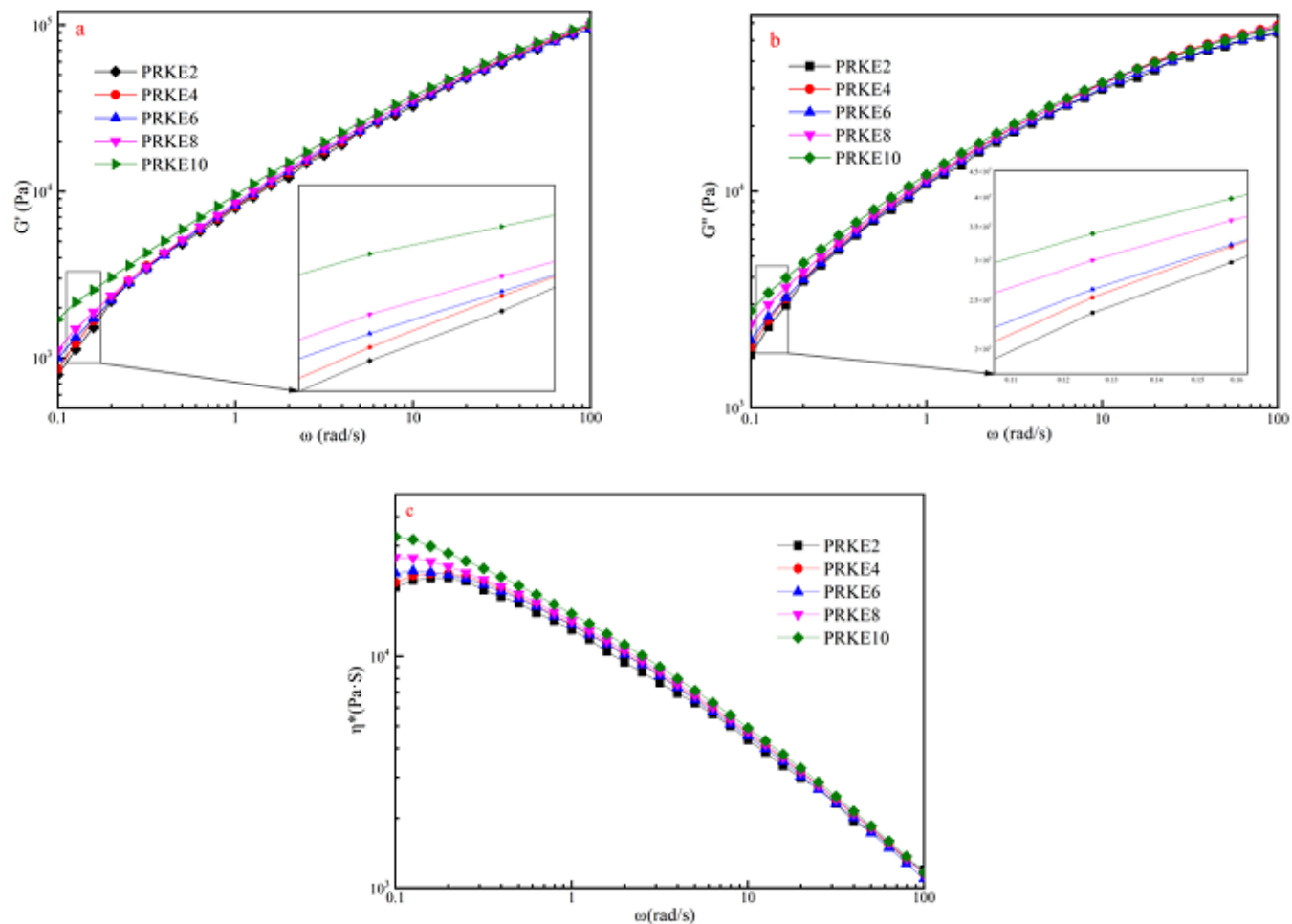


Figure 10

Rheological properties of PRKE: (a) storage modulus; (b) loss modulus;
(c) complex viscosity

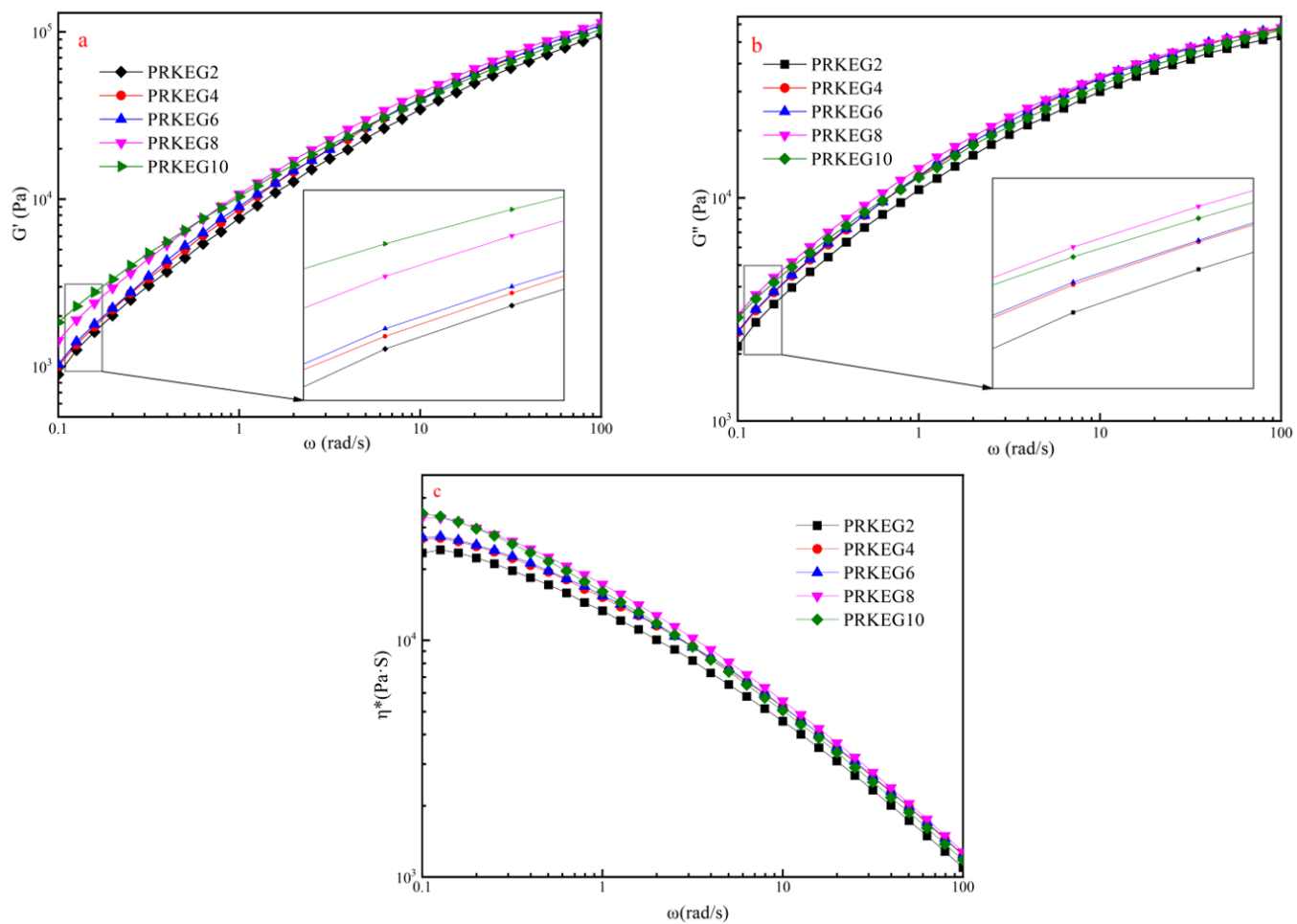


Figure 11

Rheological properties of PRKEG: (a) storage modulus; (b) loss modulus;
(c) complex viscosity

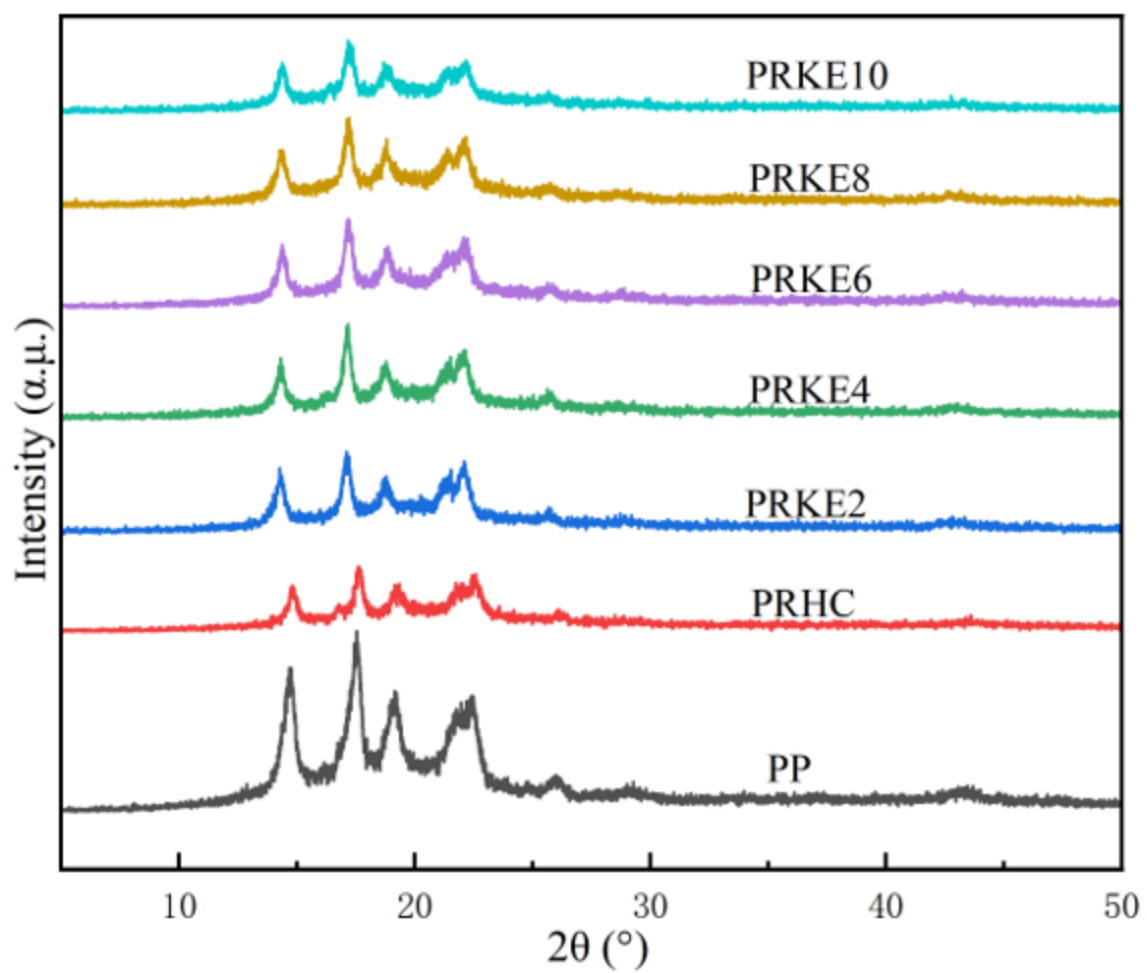


Figure 12

The XRD curve of the PP composites

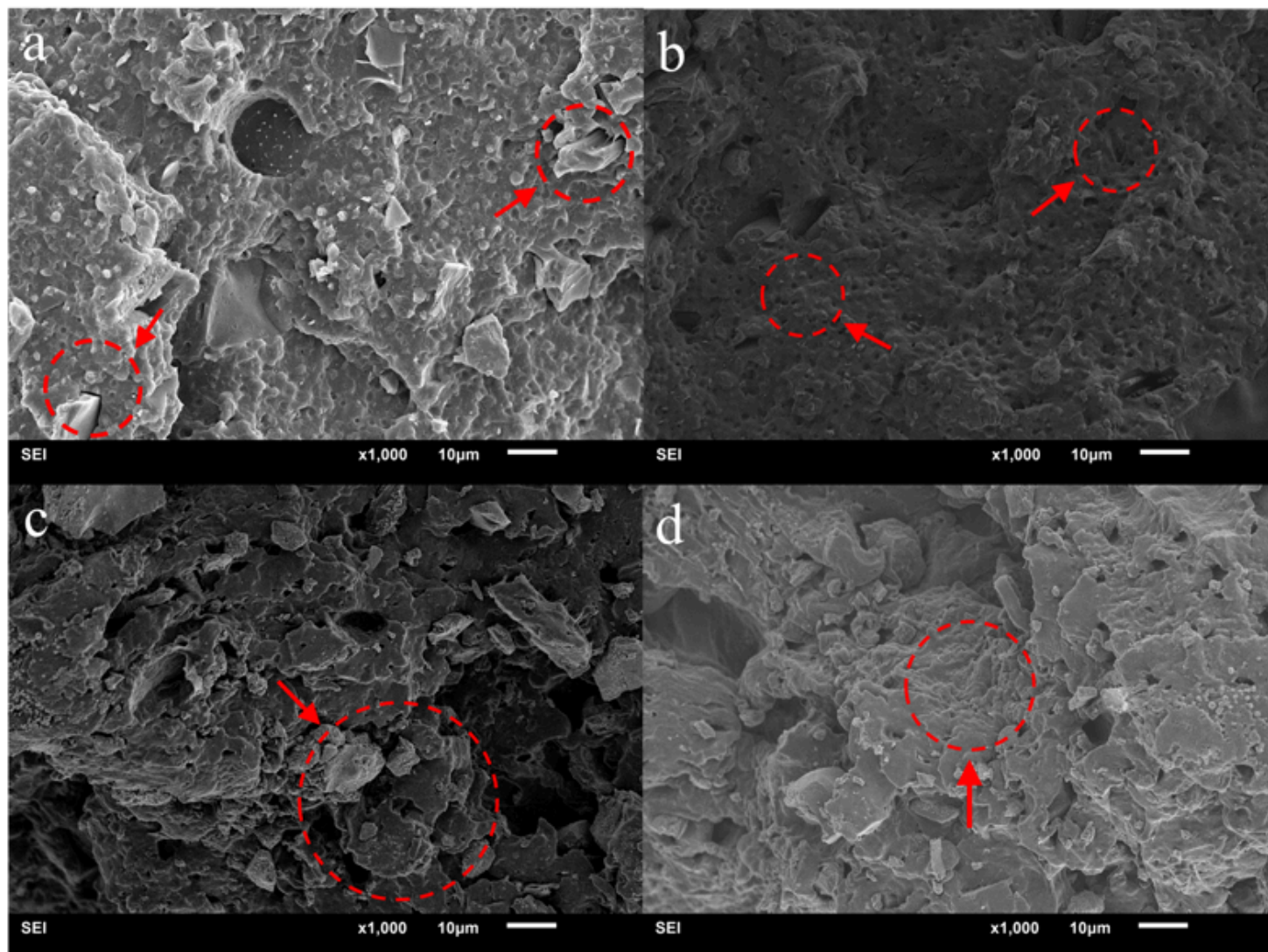


Figure 13

SEM of PP composites: (a) PRHC; (b) PRKEG4; (c) PRKEG6; (d) PRKEG10